Robustness across conditions

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

One complete research-case evidence record.

Compare frozen procedures across declared conditions without replacing slices with one pooled score.

Keep one habit: Recall → W1–W3/W5 → independent case → project evidence → feedback and delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Inspecting failures and anomalies. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Reproducible environment and records. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Predefining an analysis plan. Give an example and a mistake that the idea helps you avoid.

Inspecting failures and anomalies

> **My remembered idea, example and correction**

Reproducible environment and records

> **My remembered idea, example and correction**

Predefining an analysis plan

> **My remembered idea, example and correction**

W1 · Explain the current example

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Robustness checks compare a frozen procedure under declared varied conditions. Keep baseline, scoring and eligibility aligned and report slices separately. Main current-source pass rate is 9/10 = 90%, outdated-source 5/10 = 50%, a forty-percentage-point descriptive gap. This association does not identify a cause. Separate frozen-classifier picture has indoor 7/8 = 87.5% and outdoor 3/8 = 37.5%, gap fifty points.

  Pooling yields 10/16 = 62.5%; reporting only that value hides outdoor weakness. The observed case mix need not match future use. Eight cases per slice do not establish stable population rates, significance or universal robustness. Aggregate counts cannot reconstruct individual paired transitions. New conditions extend the evidence only to their tested scope.

  The code calculates supplied summaries, without running a model, collecting cases or verifying actual controls.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

W2 · Trace the worked case

> Keep the supplied inputs, comparison rule and evidence together.

Complete task and evidence

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  The code calculates supplied summaries, without running a model, collecting cases or verifying actual controls.

- Method passes 9/10 on current sources and 5/10 on outdated sources. Show the intermediate steps and give the conclusion with its scope.

Main quantity	Current sources	Outdated sources
Passes / evaluated	9/10	5/10
Observed rate	90%	50%
Current minus outdated	40 percentage points	Lower quality associated with this condition here
Mechanism	Not identified	Inspect source validity and procedure traces
Generalization	Only supplied cases	No universal causal or robustness conclusion

My reasoning and evidence

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task’s inputs,rule and response together.

Complete task and evidence

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  The code calculates supplied summaries, without running a model, collecting cases or verifying actual controls.

- A report omits this qualification: Robustness is bounded by the tested variations and independent **evidence** policy. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Robustness checks compare a frozen procedure under declared varied conditions. Keep baseline, scoring and eligibility aligned and report slices separately. Main current-source pass rate is 9/10 = 90%, outdated-source 5/10 = 50%, a forty-percentage-point descriptive gap. This association does not identify a cause. Separate frozen-classifier picture has indoor 7/8 = 87.5% and outdoor 3/8 = 37.5%, gap fifty points.

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  The code calculates supplied summaries, without running a model, collecting cases or verifying actual controls.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Baseline/candidate on condition A:6/8, 7/8; condition B:5/8, 4/8. Compare paired rates.
- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

My independent explanation, trace and limits

MAIN LESSON · SECTION 10

Project · Keep the research evidence

> Preserve the question, protocol, observations, analysis and limits.

Complete task and evidence

- Add a labelled robustness across conditions **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of robustness across conditions.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for Robustness across conditions. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for Robustness across conditions. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Baseline/candidate on condition A:6/8, 7/8; condition B:5/8, 4/8. Compare paired rates.

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

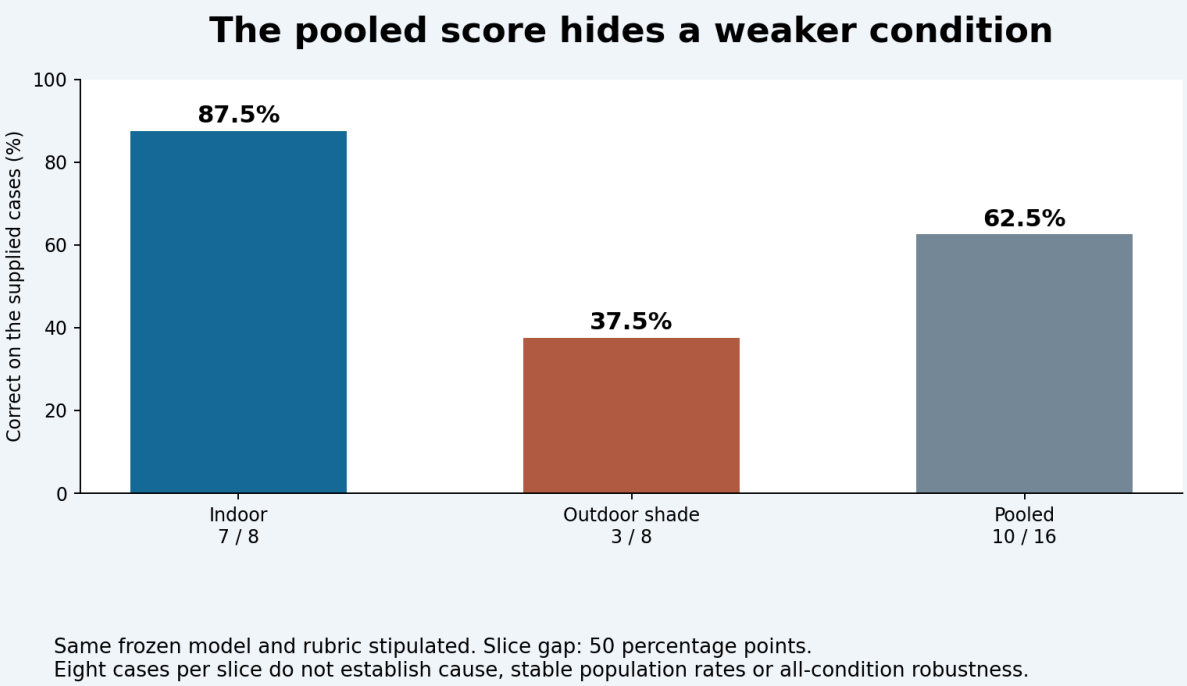
> Keep this task's inputs,rule and response together.

Complete task and evidence

- Check robustness across conditions. A fixed classifier is tested separately on indoor and shaded outdoor cases.
- Indoor 7/8 correct; outdoor 3/8 correct. The rubric and frozen model are unchanged.
- Calculate indoor, outdoor and combined rates. Which relevant weakness disappears from view when only the combined rate is reported?
- Support used: alone / hint / worked example. This record is practice **evidence**.

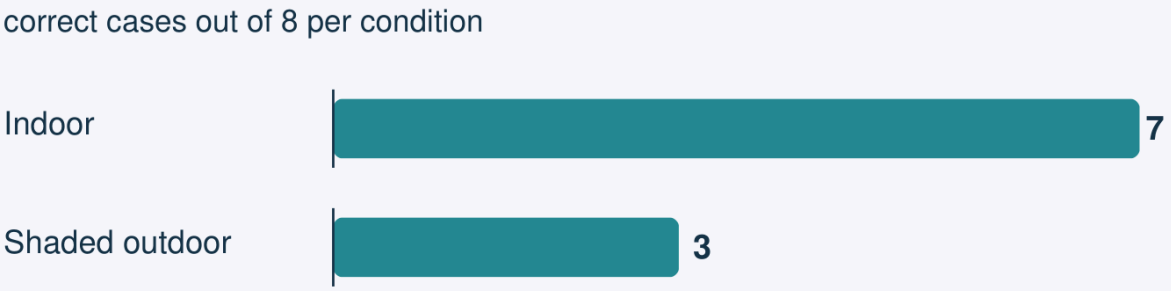
- Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

  Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.



AI ACADEMY / AI-7/27

Fixed model across conditions



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/27

Robustness across conditions

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Name both slices

Each condition has eight labelled cases.

2. Compute rates

Indoor is 87.5%; outdoor is 37.5%.



3. Bound robustness

The large observed gap motivates investigation rather than claiming condition independence.

CHANGE ONE THING

Report only the combined 10/16=62.5%.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Project: Reproducible AI Research Study
- Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.
- For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep	My record
This week’s input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Check robustness across conditions. A fixed classifier is tested separately on indoor and shaded outdoor cases.
- Which labelled step matches this detail? The large observed gap motivates investigation rather than claiming condition independence.
- 1. Name both slices
- 2. Compute rates
- 3. Bound robustness
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Bound robustness The large observed gap motivates investigation rather than claiming condition independence.

Robustness asks whether behaviour survives relevant changes in conditions. A total score can hide a weak slice; keep the scenario and denominator visible.

Changed-case question and optional hint

- Changed-case practice: Report only the combined 10/16=62.5%. Which conclusion is supported by the supplied facts?
- 1. The arithmetic is correct, but the condition gap becomes hidden.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: The arithmetic is correct, but the condition gap becomes hidden. Combining records discards the slice distinction relevant to the deployment goal.

Repair and retain the record

Review Inspecting failures and anomalies (ai-7/26). Goal: Explain the **mechanism** and terminology of inspecting failures and anomalies. Short practice: Calculate mean and median for 2,2,2,10.

What does the timeout log establish, and why is deleting that inconvenient record incompatible with the stated plan?. Return to this week and explain what you changed.

Review Reproducible environment and records (ai-7/24). Goal: Explain the **mechanism** and terminology of reproducible environment and records. Short practice: Trace the recorded mean for 2,4,6 and its changed result for 2,4,9.

Which artefacts must accompany the **output** so another team can identify the exact run?. Return to this week and explain what you changed.

Review Predefining an analysis plan (ai-7/21). Goal: Explain the **mechanism** and terminology of predefining an analysis plan. Short practice: Apply the recorded exclusion rule.

Can two readable model failures be removed, and how must a necessary later plan change be disclosed?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

▼ Reference · Six-picture response sheets

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Robustness across conditions: the case

AI ACADEMY / AI-7/27 · 01 · LOOK AT THE FACTS

Robustness across conditions: the case

The situation

Check robustness across conditions. A fixed classifier is tested separately on indoor and shaded outdoor cases.

Supplied evidence

Indoor 7/8 correct; outdoor 3/8 correct. The rubric and frozen model are unchanged.

Explain robustness across conditions using the records and boundaries of this fictional investigation.

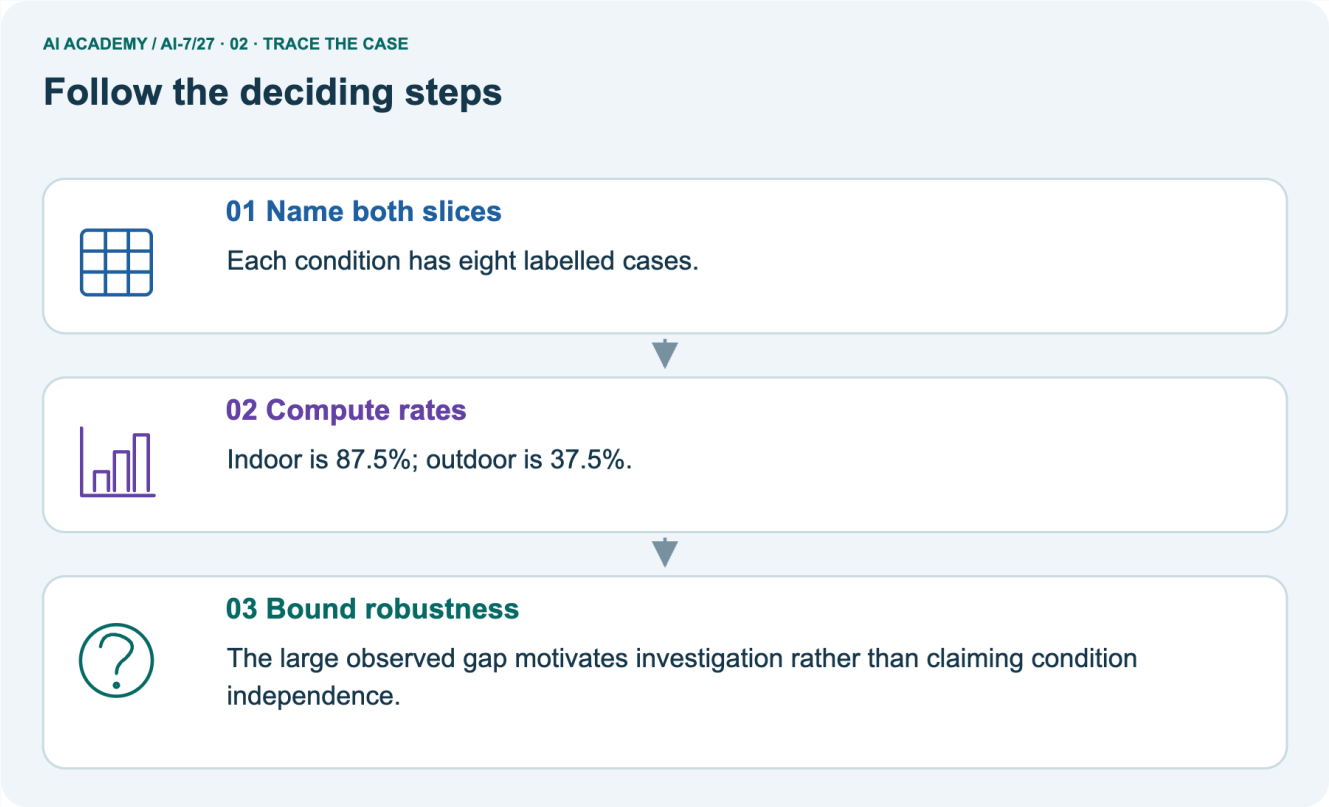
Follow the labelled picture with the matching explanation. Open the vector diagram

Check robustness across conditions. A fixed classifier is tested separately on indoor and shaded outdoor cases. Supplied facts: Indoor 7/8 correct; outdoor 3/8 correct. The rubric and frozen model are unchanged.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



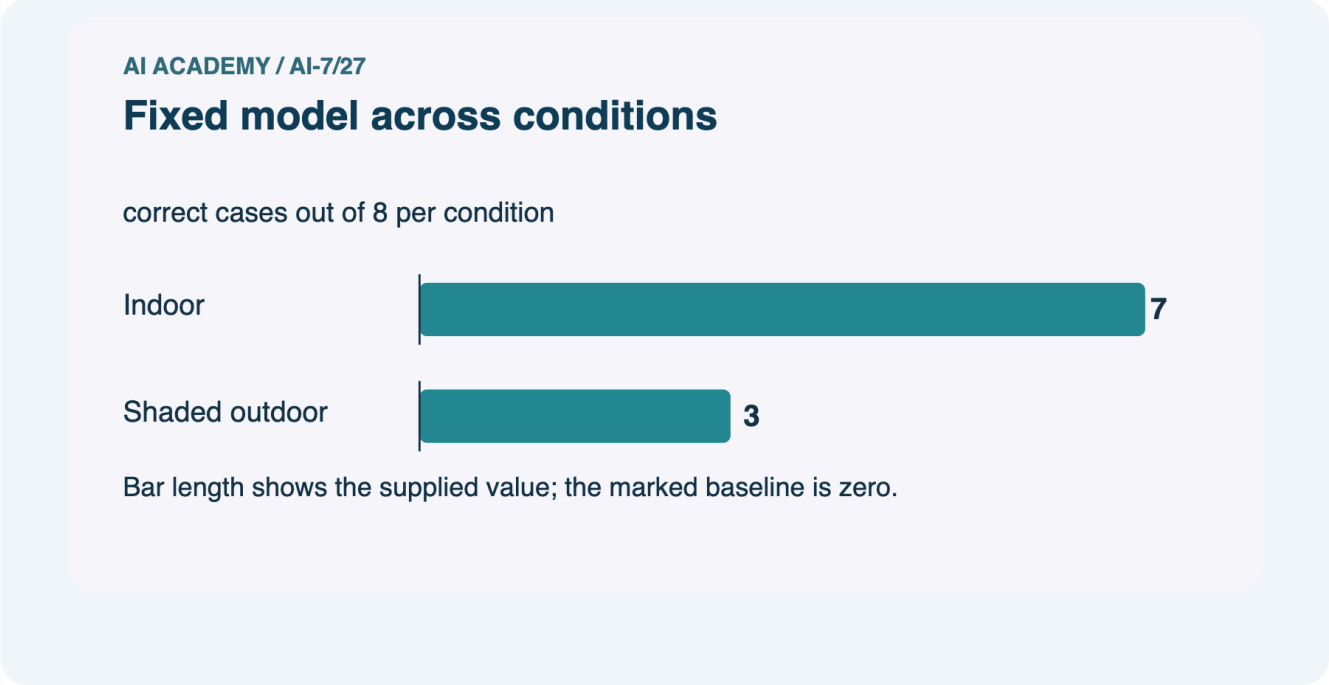
Follow the labelled picture with the matching explanation. Open the vector diagram

Name both slices: Each condition has eight labelled cases. → Compute rates: Indoor is 87.5%; outdoor is 37.5%. → Bound robustness: The large observed gap motivates investigation rather than claiming condition independence.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Fixed model across conditions



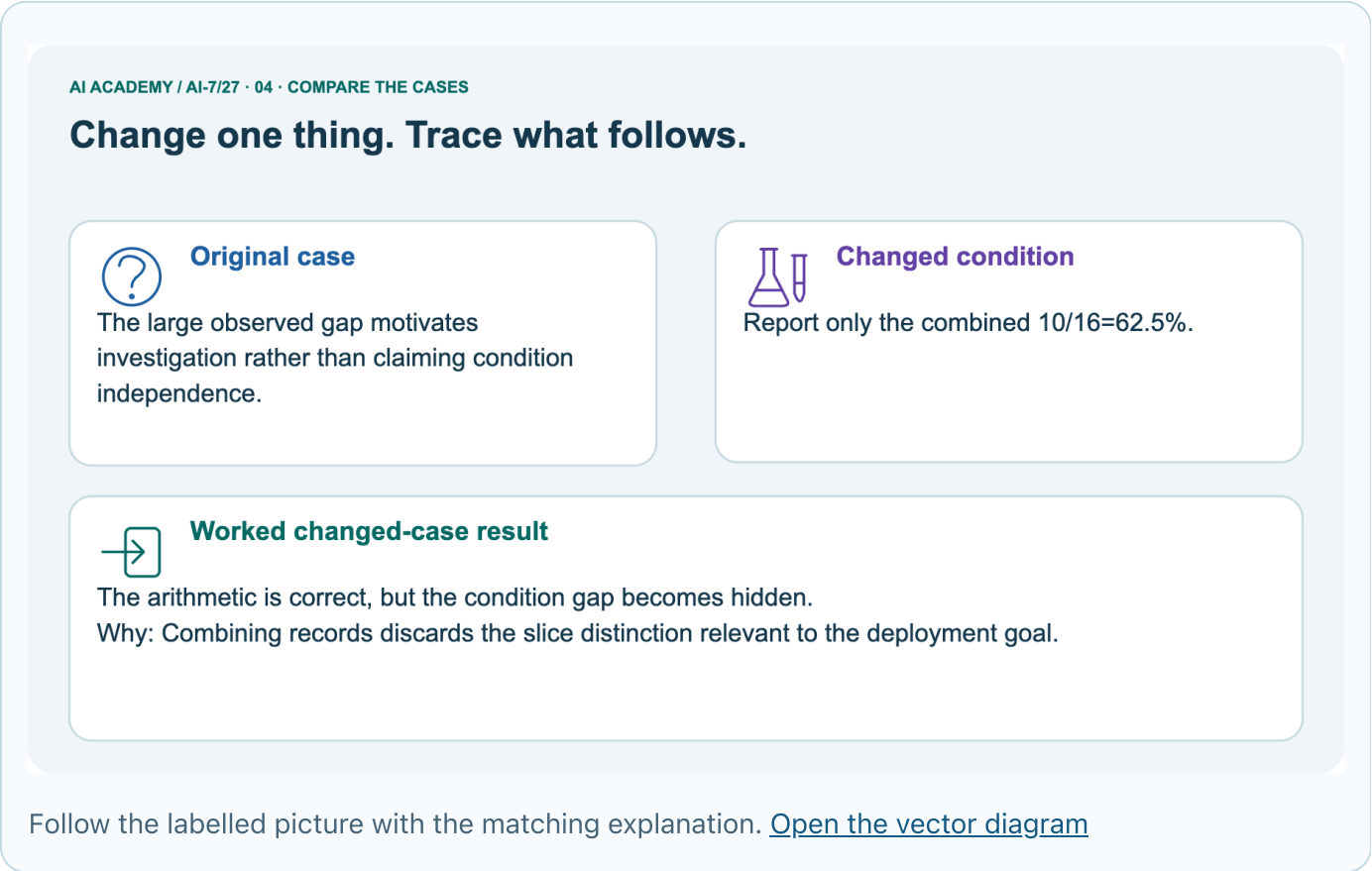
Follow the labelled picture with the matching explanation. Open the vector diagram

Fixed model across conditions. Units: correct cases out of 8 per condition. Indoor: 7. Shaded outdoor: 3.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

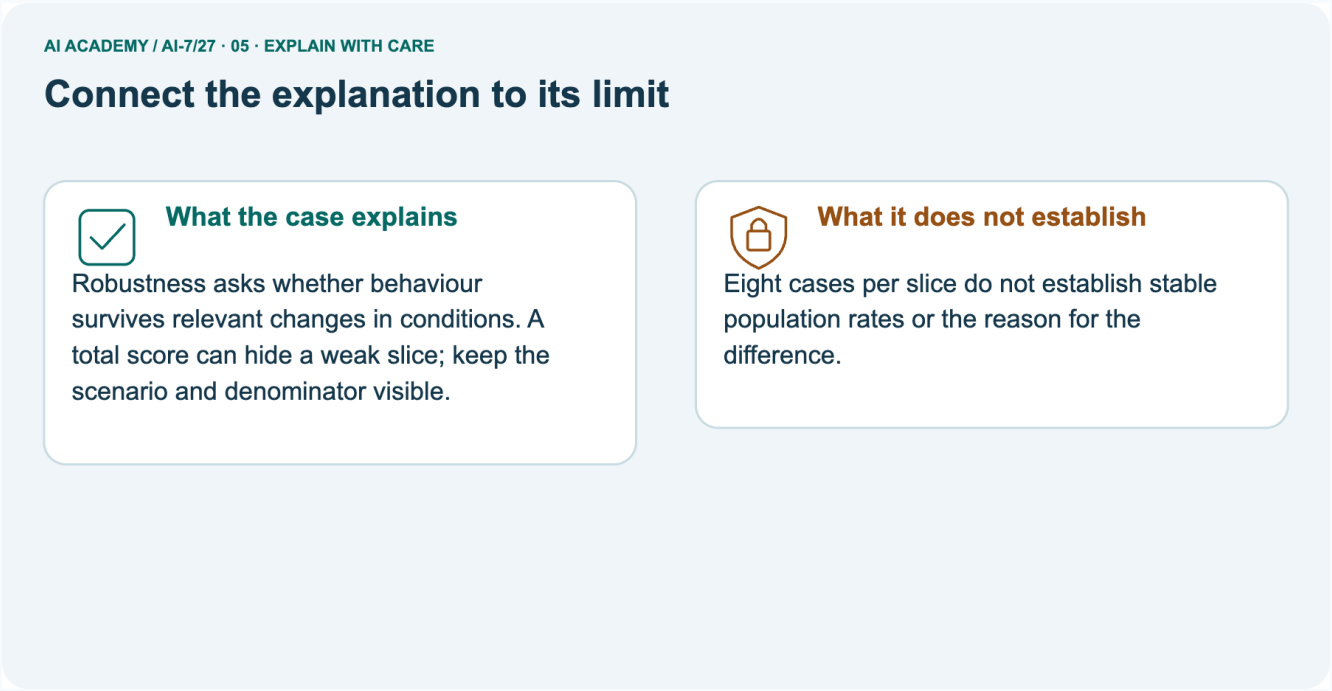
Original case: The large observed gap motivates investigation rather than claiming condition independence. Changed condition: Report only the combined 10/16=62.5%.

Expected result: The arithmetic is correct, but the condition gap becomes hidden. Reason: Combining records discards the slice distinction relevant to the deployment goal.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

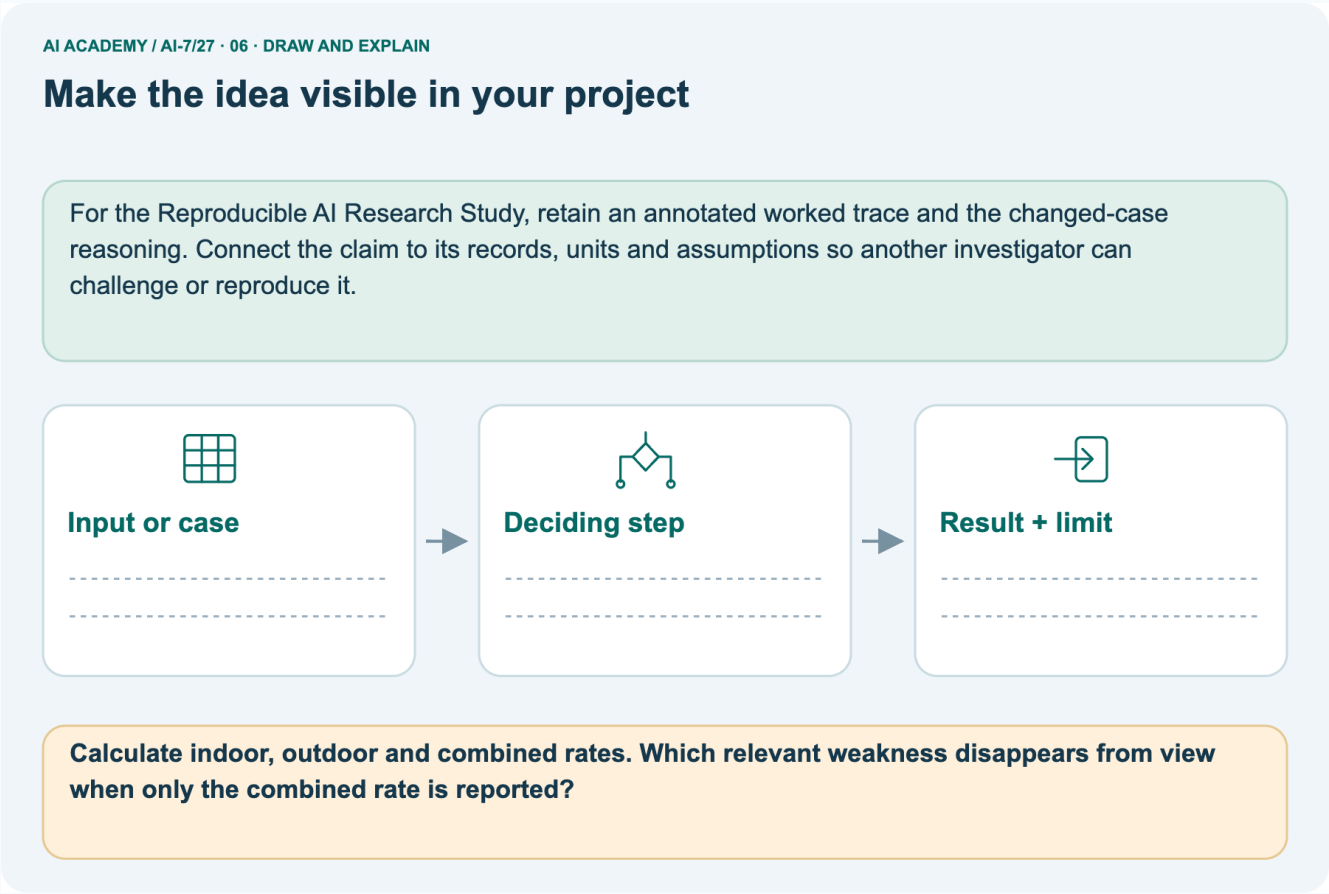
Robustness asks whether behaviour survives relevant changes in conditions. A total score can hide a weak slice; keep the scenario and denominator visible.

Limit: Eight cases per slice do not establish stable population rates or the reason for the difference.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate indoor, outdoor and combined rates. Which relevant weakness disappears from view when only the combined rate is reported?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Robustness across conditions: the case

02. Follow the deciding steps

03. Fixed model across conditions

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.